

$$I = 7.81 \text{ A}$$

- 26 D** The coil is connected to an ammeter (a closed circuit), so there is induced current in the circuit. By Lenz's law, the magnetic force acting on the induced current should oppose the rotation of the coil. Hence, the rotation will slow down, the period of rotation will increase. So, the answer is either B or D.

According to Faraday's law, the emf induced is proportional to the rate of change of the flux linkage. If the rotation slows down, the rate of change of the flux linkage decreases, hence the amplitude of the graph will decrease. The answer is **D**.

Mathematically,

$$\text{emf} = -\frac{d\Phi}{dt} = -NBA \frac{d}{dt} \cos \omega t = NBA\omega \sin \omega t$$

As the rotation of the coil slows down, the period T increases and $\omega = 2\pi f = \frac{2\pi}{T}$ (hence the amplitude $NBA\omega$) decreases.

Note: The magnetic force damps the rotation of the coil. But take note of the difference between this damping and the other examples of damping of SHM, such as a spring-mass system in water. For a spring-mass system in water, its amplitude and period of oscillation are unrelated. When damped, its amplitude will decrease exponentially in time, but its period (or frequency) can be taken to be unchanged.

- 27 A** In the initial orientation, the component of B perpendicular to the area is